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Cascades: A platform for power system security analyses and transmission expansion planning

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Software

- Review
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Summary

Cascades is an open-source MATLAB platform for power system security analyses and transmission expansion planning. It is built around a cascading failure simulation (CFS) model that supports single-zone and multi-zonal studies of interconnected transmission systems. The CFS model simulates the propagation of an initiating contingency comprising one or more simultaneous component failures, identifies electrically separated islands, balances generation and demand, applies emergency control actions, recomputes power flows, and applies configured branch-tripping rules. Users can choose AC or DC power-flow formulations and either a single control zone or multiple interconnected zones. Operational responses include reserve-based frequency control and under-frequency load shedding. AC simulations also support under-voltage load shedding and voltage regulation by on-load tap-changing (OLTC) transformers. Outputs include demand not served, cascade stages, statistics on branch overloads and outages, AC voltage violations, reserve utilization, and risk and resilience metrics. CFS results support model calibration and transmission expansion planning. The repository includes setup instructions, run scripts, and example power systems that users can run and adapt to project-specific grid models (Gjorgiev et al., 2026b, 2026a; Gjorgiev, 2026).

Statement of need

Conventional power-system security assessment relies heavily on deterministic contingency criteria such as N-1 screening. These criteria remain widely used in planning and operations. However, a single post-contingency calculation does not represent island formation, frequency imbalance, voltage deterioration, and further branch trips that can follow a sequence of dependent outages. Risk-informed planning also requires repeated simulations across operating conditions and contingencies, consistent outcome measures, and a way to relate those outcomes to candidate grid reinforcements. NERC TPL-001-5.1 requires evaluation of specified multiple-contingency planning events and extreme events. If an extreme-event assessment identifies cascading, possible actions to reduce its likelihood or mitigate its consequences must also be evaluated (NERC, 2020). Similarly, ENTSO-E's emerging operational probabilistic risk-assessment methodology complements N-1 analysis by combining contingency probabilities with impact assessment, creating a need for cascading-failure models that quantify the risk of multistage outages (ENTSO-E, 2025).

For interconnected systems, cascading-failure analysis must distinguish control zones to represent their operations. Each zone has its own generation, demand, scheduled exchanges, reserves, and balancing actions. Outages can also split or merge zones. A single aggregated control area cannot represent these processes (Stankovski et al., 2022). Severe cascades are rare, and simulation results depend on uncertain failure probabilities and protection settings. Calibration can therefore be used to compare simulated distributions of demand not served and branch outages with historical or benchmark statistics (David et al., 2020; Gjorgiev et al., 2019).

Cascades combines cascading-failure simulation, model calibration, and transmission expansion planning in one research workflow. It allows power-system researchers and transmission-planning analysts to rank branches by their impact on demand not served and their overload frequency, assess reserves and emergency controls, and test whether candidate network investments reduce cascade risk (Gjorgiev, David, et al., 2022; Gjorgiev & Sansavini, 2022; Stankovski et al., 2022). Cascades is a steady-state model and does not simulate electromechanical transients such as generator swing dynamics (Gjorgiev et al., 2026a).

State of the field

MATPOWER provides widely used steady-state AC and DC power-flow and optimal-power-flow routines (Zimmerman et al., 2011). MATCASC is an open-source MATLAB tool for cascading line-overload and islanding analysis using a DC representation (Koç et al., 2013). AC cascading-failure models can represent reactive-power and voltage effects omitted by DC models (Noebels et al., 2022). The speed-fidelity trade-off depends on the study. A direct comparison of the AC Manchester and DC OPA models found that the AC model was approximately seven times slower and that the two models produced different risk estimates under high system stress (David et al., 2020).

Cascades uses MATPOWER-compatible network data and solvers and adds multistage cascade transitions, AC and DC implementations, single-zone and multi-zonal control logic, model calibration, and transmission expansion planning. Its contribution is this integrated workflow rather than a new general-purpose power-flow solver (Gjorgiev et al., 2026a; Gjorgiev, 2026; Stankovski et al., 2022).

Software design

The software is organized around a common cascading-failure simulation (CFS) framework with separate AC and DC implementations that use MATPOWER for power-flow and optimal-power-flow calculations (Gjorgiev, 2026; Zimmerman et al., 2011). For each operating condition and initiating contingency, the selected implementation updates the network topology, identifies islands, evaluates frequency imbalance, deploys available reserves, applies configured emergency controls, and solves the resulting power-flow problem. The AC implementation also evaluates bus voltages. Branch flows are then checked against configured tripping thresholds. In the standard deterministic mode, the most overloaded branch is tripped at each stage. For islands classified as blacked out, demand and power flows are set to zero; surviving islands continue until no branch-flow violation remains or the entire system has collapsed. Post-processing reports demand not served, branch outages and overloads, AC voltage violations, and reserve utilization. The overall platform organization is shown in Figure 1.

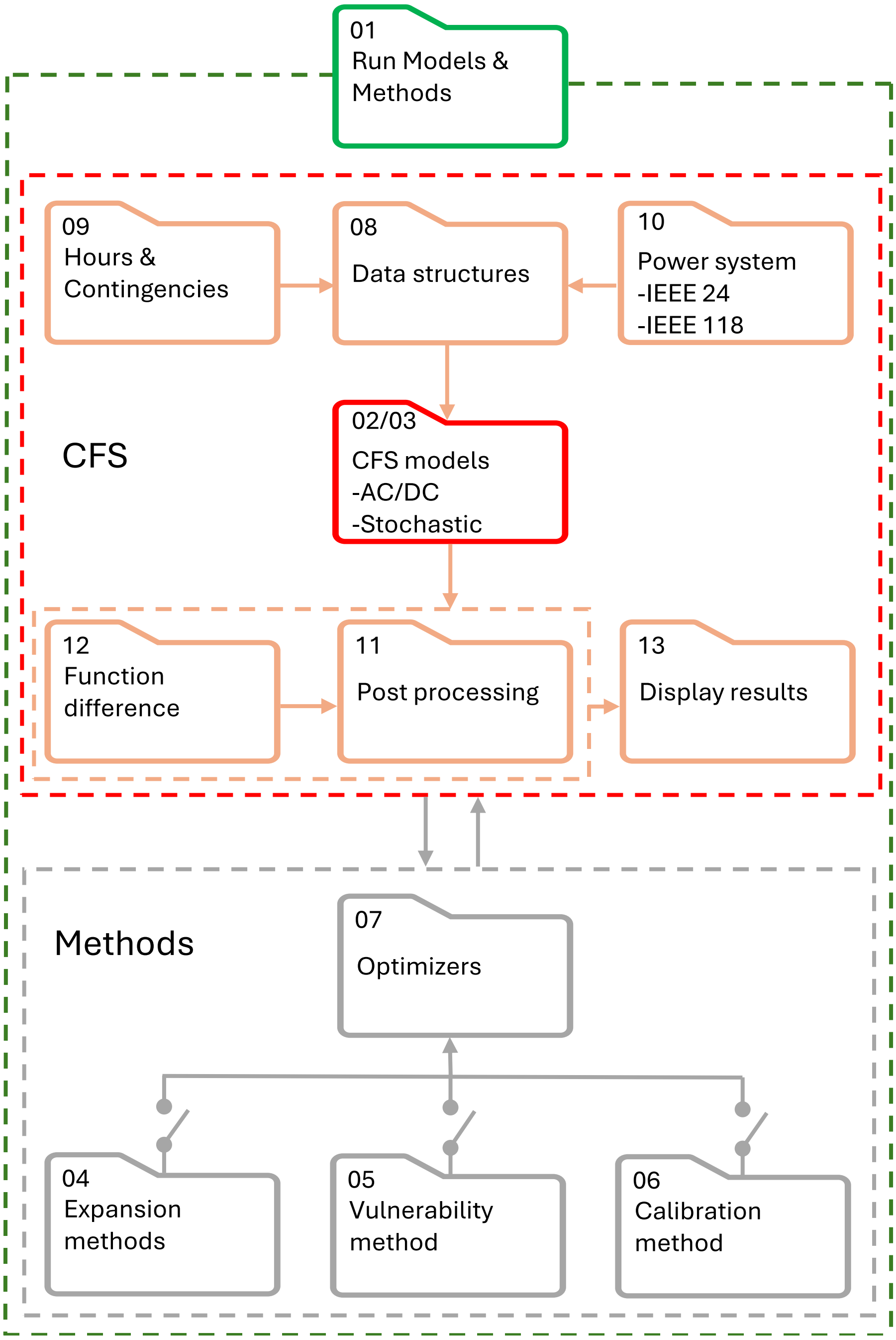

Figure 1: Functional dependencies among the Cascades algorithm folders. The CFS block contains the AC/DC core and its input, data-processing, post-processing, comparison, and display components. Expansion planning and calibration use the CFS to evaluate an objective function value within a dedicated optimizer. Folder 03 is reserved for future development of the fully stochastic CFS, and folder 05 is a placeholder because its vulnerability-analysis implementation is not publicly distributed. The figure is adapted from Figure 2 of the Cascades user manual (Gjorgiev, 2026).

The AC and DC implementations share orchestration and data-handling functions but retain formulation-specific solution and correction routines. The DC implementation omits reactive-power and bus-voltage calculations, making it computationally simpler but unable to represent voltage-dependent actions. The AC implementation represents losses, reactive-power limits, and voltage behavior and offers Newton-Raphson, Gauss-Seidel, and two fast-decoupled power-flow algorithms. Configurable remedies for non-convergent post-contingency states include stepwise reductions in load and generation, full AC optimal power flow, and reactive-power redispatch; if these measures fail, the model can continue from the non-converged state or treat it as a blackout (Gjorgiev et al., 2026a; Gjorgiev, 2026). In multi-zonal simulations, control-zone topology and scheduled exchanges are updated as lines trip and zones split or merge. Frequency control and reserves are applied per zone; if reserves are insufficient, the model can reduce scheduled exports and shed load, and it can redispatch generation to alleviate overloads introduced by frequency control (Stankovski et al., 2022).

`Cascades` represents corrective and protection mechanisms within the cascade loop. Frequency control changes the output of synchronized generators within their ramp-rate and operating limits, while under-frequency load shedding (UFLS) disconnects load according to configured frequency thresholds. In AC simulations, under-voltage load shedding (UVLS) removes load stepwise at an undervoltage bus and recalculates the power flow until the voltage recovers or the load is exhausted. Optional on-load tap-changing (OLTC) control compares regulated-bus voltages with setpoints and deadbands, changes transformer taps in discrete steps within their limits, and reruns the AC power flow. Branch tripping can use deterministic or stochastic overload thresholds, and an optional hidden-failure mechanism can trip additional connected branches. After each branch trip, the topology is updated and the control calculations are repeated (Gjorgiev et al., 2026a; Gjorgiev, 2026).

`Cascades` loads MATPOWER-compatible case data into an internal `data` structure augmented with reliability parameters, reserve schedules, control-zone assignments, candidate-branch data, and optional OLTC fields. Existing MATPOWER cases can therefore be used after the additional inputs required for the chosen study are supplied. The expansion-planning modules use the CFS framework in two ways: (1) optimization-based searches implemented with a genetic algorithm, the Non-dominated Sorting Genetic Algorithm II (NSGA-II), or Jaya, and (2) importance-list methods derived from simulated branch-failure impacts and overload frequencies (Gjorgiev, David, et al., 2022; Raycheva et al., 2023). The calibration module uses repeated CFS runs to tune branch-tripping thresholds and branch failure probabilities against reference outage statistics (Gjorgiev et al., 2019).

`Cascades` can run either in standalone mode or through an interface to another model. Standalone runs load MATPOWER-compatible cases and time-series inputs from local files, while the interface mode exchanges data with external models, as demonstrated in Nexus-e (Gjorgiev, Garrison, et al., 2022). The `configStandAlone.m` and `configNexuse.m` scripts specify the settings for these two modes. The `configSampleRun.m` script specifies selected operating conditions and contingencies for inspecting model behavior and debugging edge cases before larger Monte Carlo or optimization studies (Gjorgiev et al., 2026b; Gjorgiev, 2026).

For standalone studies, `Cascades` reads system data from a subfolder of `10_Power_system`; the code also provides a MySQL input route. Each local system contains a `System` file in one of three formats (`.m`, `.xlsx`, or `.ods`) that supplies `baseMVA`, `bus`, `gen`, `branch`, and `gencost` data in MATPOWER `mpc` format. Other run inputs include nominal system frequency, time-dependent nodal demand, and, where applicable, non-dispatchable generation profiles. Documented fallback routines can construct some inputs when detailed data are unavailable. Study-specific inputs can provide generation dispatch, reserves, branch and generator failure probabilities, under-frequency load-shedding schemes, and candidate branches. The repository includes IEEE- and PEGASE-based cases. The `IEEE118` case is the recommended starting point because it contains the most complete example dataset (Gjorgiev, 2026).

Research impact statement

`Cascades` and its underlying methods have been used in published studies of transmission-asset vulnerability (Gjorgiev & Sansavini, 2022), multi-zonal cascade behavior (Stankovski et al., 2022), cascade-risk-informed transmission expansion (Gjorgiev, David, et al., 2022), and coordinated generation and transmission expansion (Raycheva et al., 2023). The vulnerability study coupled the CFS framework to a study-specific optimization layer that is not included in this public release. The calibration method incorporated into `Cascades` was reported by Gjorgiev et al. (2019), and `Cascades` served as the network-security and expansion module in Nexus-e (Gjorgiev, Garrison, et al., 2022).

The multi-zonal study used the platform to update control-zone topology and apply zone-specific balancing as cascades evolved (Stankovski et al., 2022). The expansion studies used CFS-derived risk curves to evaluate candidate reinforcements (Gjorgiev, David, et al., 2022). The Nexus-e interface passed demand, generation dispatch, and reserve data to `Cascades` and returned proposed branch upgrades (Gjorgiev, Garrison, et al., 2022; Raycheva et al., 2023).

`Cascades`-generated scenarios have also supported graph-based machine learning. They were used to train a graph neural network classifier that determines whether specified outages under given operating conditions lead to a critical grid state (Varbella et al., 2023). `Cascades` also generated the cascading-failure component of the PowerGraph benchmark datasets (Varbella et al., 2024). In later work, a graph neural network regression model trained on `Cascades` outputs was used to evaluate cascading-failure risk in goal-oriented graph generation for transmission expansion planning (Varbella et al., 2025).

AI usage disclosure

The software and its principal technical documentation were produced by human developers. Nevertheless, OpenAI ChatGPT and Codex (GPT-5.3, GPT-5.4, and GPT-5.5) were used to a limited extent for consultation and later software maintenance, including in building the OLTC functionality. Finally, the authors used OpenAI Codex (GPT-5.6) to review the repository against JOSS requirements, draft tests, and assist with manuscript editing. The authors accept full responsibility for the accuracy, originality, licensing, and scientific content of the software, documentation, and paper.

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